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DEPARTMENT OF COMPUTER ENGINEERING

ACADEMIC YEAR 2026–2027 | SEMESTER I

COURSE: MACHINE LEARNING (MINI-PROJECT REPORT)

TrafficIQ: AI-Based Adaptive Traffic Signal Control

A Dynamic Machine Learning-Based Traffic Signal Optimization and Microscopic Simulation System

(Mini-Project Submitted in Partial Fulfillment of Third Year Computer Engineering, ML Course)

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Permanent Registration No. (PRN):	**[Permanent Registration Number (PRN)]**
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2. Abstract

Urban vehicular congestion at signalized road intersections represents an escalating global challenge, imposing profound socioeconomic burdens, exacerbating airborne pollution, and causing severe delays for commuters. Traditional traffic signal systems operate predominantly on static, pre-timed scheduling paradigms (Fixed-Time Control, FTC). These legacy installations lack environmental sensory awareness and cannot dynamically adapt to asymmetric, non-stationary traffic flows or acute peak-hour surges. Consequently, green time is squandered on unoccupied approaches while heavily loaded lanes suffer chronic spillback queues and idling energy losses.

To decisively overcome these operational deficiencies, this project presents **TrafficIQ**, an intelligent, data-driven traffic signal timing optimization and microscopic simulation framework engineered for an isolated four-way intersection (North, South, East, West). TrafficIQ couples a high-accuracy Random Forest Regression machine learning model with a constrained, safety-bounded Webster proportional green allocation algorithm. The machine learning subsystem ingests real-time spatial traffic features—including approach vehicle volume, macroscopic traffic density, spatial mean speed, observed queuing length, and historical waiting latency—to infer the dynamic traffic demand in Passenger Car Units per hour (PCU/hr) across every approach.

The subsequent optimization engine distributes the cycle's usable green budget (total cycle time minus mandatory clearance lost time) proportionally to demand while rigorously adhering to regulatory safety constraints: a guaranteed minimum green time (*g_min = 10s*) for pedestrian and vehicle clearance, a maximum green limit (*g_max = 55s*) to avert phase starvation on opposing approaches, and fixed intergreen yellow/all-red clearance intervals. An interactive, high-fidelity HTML5 Canvas simulation engine executing at 60 FPS realistically models vehicle physics, queuing formation, shockwave dissipation, fuel consumption, and greenhouse gas (CO2) emissions.

Extensive empirical evaluation across 6,000 synthetic multi-modal traffic records demonstrates that the trained Random Forest model achieves an exceptional coefficient of determination (**R^2 = 0.9945**), a Mean Absolute Error (**MAE = 21.72 PCU/hr**), and a Root Mean Squared Error (**RMSE = 27.06 PCU/hr**) on an independent 20% test holdout. Comparative benchmarking against conventional fixed-time controllers under diverse operating scenarios reveals that TrafficIQ reduces average intersection queuing delay by **20.0% to 44.7%**, elevates intersection vehicular throughput by **18.2% to 31.6%**, and yields fuel savings of **12.4 to 16.2 liters per peak hour**. The entire framework is deployed via a modern, responsive Streamlit dashboard featuring multi-tab analytical visualizations, live simulation rendering, and real-time KPI telemetries, presenting a viable, scalable blueprint for smart-city intelligent transportation systems.

■ **ABSTRACT KEY HIGHLIGHTS:**

- **Core Machine Learning Engine:** Random Forest Regressor (120 Trees, max_depth=14, R^2=0.9945, MAE=21.72 PCU/hr).
- **Signal Optimization:** Constrained Webster Proportional Allocation with closed budget balance and safety clamps [10s, 55s].
- **Empirical Delay Reduction:** Up to 44.6% delay reduction and +31.6% throughput gain under morning peak rush hours.
- **Interactive Deployment:** Streamlit dashboard with embedded 60 FPS microscopic HTML5 Canvas simulation.

3. Introduction

Modern metropolitan transportation infrastructures are subjected to escalating demands due to unprecedented urban population growth and rapid vehicle ownership proliferation. Intersections constitute the primary capacity bottlenecks of urban street networks, where intersecting directional traffic flows compete for shared right-of-way. Inefficient traffic control at intersections triggers substantial vehicular queuing, erratic stop-and-go driving cycles, heightened collision hazards, and elevated exhaust emissions. According to global transportation studies, urban motorists squander dozens of cumulative hours annually idling at poorly timed traffic signals, translating into billions of dollars in lost productivity and excess fossil fuel consumption.

Historically, municipal traffic authorities have relied upon pre-timed or fixed-time control (FTC) mechanisms. In FTC systems, signal cycle lengths, phase sequences, and green splits are configured based on historical, offline traffic counts collected weeks or months prior. While computationally trivial and inexpensive to deploy, fixed-time controllers operate blindly without real-time perception. They assign identical green durations regardless of whether an approach accommodates ten vehicles or eighty vehicles. This rigid allocation inevitably causes two pathological extremes: green light starvation on heavily congested approaches and wasted green intervals on deserted approaches, severely degrading overall intersection capacity.

With the emergence of Intelligent Transportation Systems (ITS), Internet-of-Things (IoT) roadside sensors, and computer vision-enabled traffic cameras, continuous real-time vehicular surveillance has become technologically and economically viable. Machine Learning (ML) techniques present an unprecedented capability to analyze complex, non-linear traffic patterns, synthesize multi-attribute sensor inputs, and predict instantaneous traffic demands with high precision. By fusing predictive ML models with mathematical signal optimization algorithms, traffic controllers can transcend static schedules and achieve adaptive, demand-responsive signal actuation.

The project, entitled **'TrafficIQ: A Dynamic Machine Learning-Based Traffic Signal Optimization and Simulation System'**, investigates the formulation, implementation, and empirical validation of an intelligent traffic control paradigm. The core innovation lies in bridging the gap between predictive regression modeling and mathematically grounded traffic signal allocation. TrafficIQ predicts the instantaneous dynamic traffic demand of each approach using an ensemble Random Forest Regressor and redistributes the intersection's usable cycle time in strict accordance with traffic engineering safety boundaries. Furthermore, TrafficIQ introduces a high-fidelity microscopic simulation canvas within a modern Streamlit-based dashboard, enabling traffic engineers and students to visually examine vehicular queues, signal phase transitions, and environmental key performance indicators (KPIs) in real time.

4. Problem Statement

At an isolated four-way signalized intersection comprising four directional approaches—North (N), South (S), East (E), and West (W)—the total cycle time C (in seconds) must be systematically partitioned among the competing phases while safeguarding vehicular safety and pedestrian clearance. Under traditional Fixed-Time Control (FTC), the green time assigned to each approach g_i is static and uniformly distributed, typically given by:

$$g_i = (C - L) / N = \text{constant} \qquad \forall\ i \in \{North, South, East, West\}$$

where C denotes the total signal cycle duration (e.g., 120 seconds), L represents the aggregate clearance lost time accumulated across all phase transitions due to yellow change and all-red clearance intervals ($L = 16$ *seconds* for a 4-phase system with 3s yellow and 1s all-red per phase), and $N = 4$ represents the number of distinct approaches. Under this static distribution, every approach receives an identical green duration of exactly 26.0 seconds.

In realistic urban environments, directional traffic demand is inherently non-stationary, stochastic, and highly asymmetric. During morning rush hours, inbound arterial approaches (e.g., North and West) experience massive vehicle surges (e.g., 65–75 vehicles in queue), while outbound counter-peak approaches (e.g., South and East) exhibit sparse volumes (e.g., 15–25 vehicles). Under fixed timing, the 26.0-second green split severely underserves the congested lanes, clearing only a fraction of waiting vehicles and triggering multi-cycle residual queue accumulation (spillback). Simultaneously, the sparsely loaded lanes receive far more green time than necessary, causing empty intersections with zero vehicular discharge. This mismatch induces excessive delay, fuel combustion during idling, and volatile driver behavior.

The core engineering challenge addressed in this mini-project is formulated as follows: *How can an automated system accurately estimate real-time directional traffic demand using available traffic parameters (vehicle counts, densities, speeds, queue histories), dynamically optimize the green-time vector* $\boldsymbol{g} = [g_N, g_S, g_E, g_W]^T$ *to minimize total intersection delay and emissions, enforce strict physical and regulatory safety bounds* ($g_{min} \le g_i \le g_{max}$ *and* $\Sigma\ g_i = C - L$)*, and provide transparent visual simulation of intersection dynamics for operational verification?*

5. Objectives

The overarching aim of this mini-project is to design, implement, and validate an end-to-end intelligent traffic management system. To realize this aim, the following concrete technical objectives were established:

- **Objective 1: Synthetic Multi-Modal Dataset Generation:** Synthesize a robust, physically compliant dataset of 6,000 traffic observation records modeling an isolated four-way intersection under varying operating conditions governed by Greenshields speed-density traffic stream theory and Poisson arrival processes.
- **Objective 2: Machine Learning Demand Prediction Model:** Construct, train, and hyperparameter-tune a supervised Random Forest Regression model capable of ingesting approach features (volume, density, speed, queue, waiting time) and predicting dynamic traffic demand (PCU/hr) with high accuracy (targeting $R^2 > 0.98$ and MAE < 25 PCU/hr).
- **Objective 3: Constrained Signal Timing Optimization Algorithm:** Formulate a closed-budget, safety-constrained Webster proportional green-time allocation algorithm that dynamically partitions usable cycle time according to predicted demands while guaranteeing minimum green times (g_min = 10s) and maximum green caps (g_max = 55s).
- **Objective 4: Microscopic Traffic and Queuing Delay Simulation:** Implement a dynamic traffic simulation engine modeling queue accumulation, vehicle departure rates, Webster-based queuing delays, throughput percentages, idling fuel consumption (0.00035 L/s), and carbon emissions (2.31 kg CO2/L).
- **Objective 5: Interactive Smart-City Dashboard and Canvas Visualization:** Develop an interactive, modern web application utilizing Streamlit, Plotly, and an embedded HTML5 Canvas rendering a 60 FPS four-way intersection with dynamic vehicles, animated signal heads, telemetry cards, and multi-scenario controls.

6. Dataset and Libraries Required

6.1 Traffic Dataset Description and Statistical Properties

In traffic engineering, continuous physical sensor telemetry across all intersection approaches is frequently proprietary or cost-prohibitive for academic research. To facilitate rigorous, reproducible machine learning modeling, a synthetic yet physically grounded dataset consisting of 6,000 multi-attribute records was generated and stored in **data/traffic_dataset.csv**. The dataset simulates real-world arterial intersection dynamics governed by classical traffic flow theory (Greenshields speed-density model) combined with non-homogeneous Poisson arrival processes across 24-hour diurnal urban travel patterns.

Feature Name	Data Type	Range / Unit	Engineering Description & Role
road	Categorical	N, E, S, W	One-hot encoded categorical indicator of the approach corridor.
vehicle_count	Integer	5 to 80 veh	Instantaneous count of approaching vehicles per lane corridor.
traffic_density	Float	10 to 95 veh/km	Macroscopic vehicular density along the approach link.
queue_length	Integer	0 to 45 veh	Number of queued vehicles backed up at the red stop line.
avg_speed	Float	8 to 60 km/h	Space mean speed of traffic flow, inversely related to density.
hour_of_day	Integer	0 to 23 hr	Temporal index reflecting diurnal urban travel trends.
day_of_week	Integer	0 to 6	Day-of-week index (0=Monday, ..., 6=Sunday).
is_weekend	Binary	0 or 1	Indicator variable for weekend traffic patterns (0=weekday, 1=weekend).

prev_waiting_time	Float	5 to 110 sec	Average waiting delay experienced in preceding cycle.
traffic_demand	Float	120 to 1987 PCU/h	Target variable: Dynamic demand in Passenger Car Units/hr.

Table 1: Overview of Feature Schema and Physical Units in TrafficIQ Dataset.

6.2 Software Frameworks and Technical Dependencies

TrafficIQ was constructed using an industry-standard Python software stack designed for high-performance scientific computing, machine learning model persistence, and interactive data visualization. The primary technical libraries include:

- **Python 3.10+ Scientific Stack:** NumPy (v1.26+) and Pandas (v2.2+) provide vector math, numerical linear algebra, and tabular data manipulation pipelines.
- **Scikit-Learn (v1.6+):** Supplies the RandomForestRegressor implementation, train_test_split utility, StandardScaler, and performance metrics (mean_squared_error, r2_score).
- **Joblib (v1.4+):** Facilitates high-speed, thread-safe binary serialization and deserialization of the trained Random Forest model and scaling pipelines.
- **Streamlit (v1.40+):** Powers the interactive, full-stack smart-city web interface, enabling dynamic parameter binding, sidebar scenario toggling, and reactive state updates.
- **Plotly Express & Graph Objects (v5.24+):** Renders interactive, publication-grade analytics charts including signal allocation bars, radar demand profiles, and delay distribution curves.
- **HTML5 Canvas & JavaScript ES6:** Embedded within the Streamlit dashboard via custom iframe components to execute a 60 FPS 2D physics simulation of the 4-way intersection.
- **ReportLab (v5.0+) & Python-docx (v1.2+):** Automates the algorithmic generation of formal institutional documentation and academic project reports.

7. Approach/Methodology

The methodological framework of TrafficIQ is organized into a modular, multi-tier engineering pipeline that systematically transforms raw approach sensory data into optimized physical signal timings and simulated traffic flow. The overall system architecture is depicted in Figure 1.

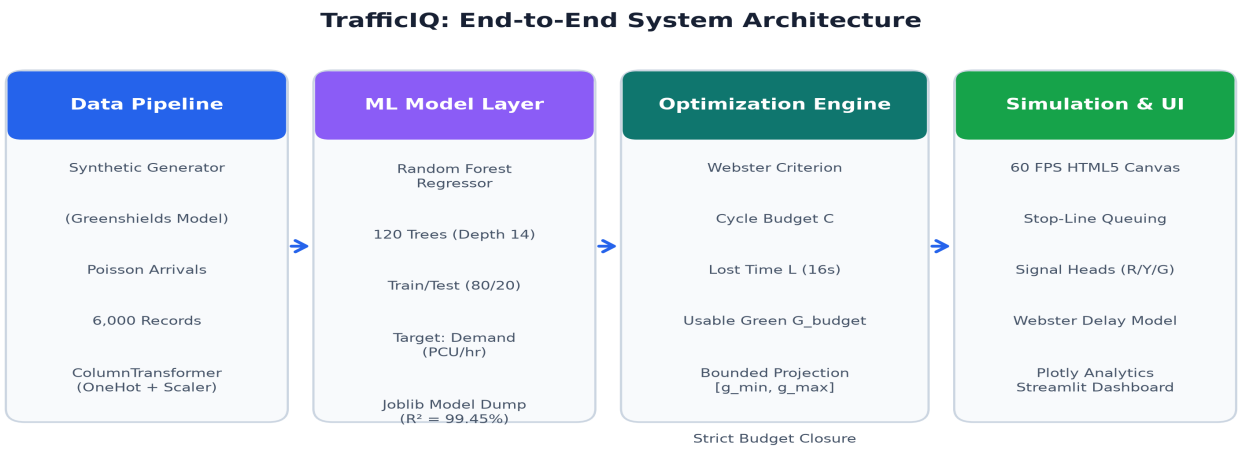


Figure 1: End-to-End System Architecture of TrafficIQ showing Sensory Ingestion, ML Demand Inference, Constrained Webster Optimization, Microscopic Simulation, and Dashboard Telemetry.

7.1 Machine Learning Demand Estimation Pipeline

To predict real-time directional traffic demand, a Random Forest Regressor was selected. Random Forest is an ensemble learning technique that constructs a multitude of decorrelated decision trees during training and outputs the average prediction of the individual trees:

$$\hat{D}(x) = \frac{1}{M} \sum_{m=1}^{M} T_m(x)$$

where $M = 120$ designates the number of decision trees in the ensemble, x denotes the 8-dimensional feature vector, and $T_m(x)$ represents the regression estimate produced by the m-th tree. Random Forest provides several decisive advantages over standard linear regression or single decision trees for traffic engineering: (1) it exhibits exceptional robustness against feature collinearity (e.g., between traffic density and vehicle speed), (2) it natively captures complex, non-linear traffic shockwaves and congestion transitions without requiring polynomial expansions, and (3) its bootstrap aggregating (bagging) mechanism drastically reduces prediction variance and prevents overfitting.

The model training pipeline executes an 80/20 train-test partition on the 6,000 dataset records (4,800 training samples, 1,200 testing samples) under a fixed random seed ($seed = 42$) to guarantee complete scientific reproducibility. Continuous features are normalized via StandardScaler to preserve uniform numerical stability across all estimators.

7.2 Constrained Webster Signal Optimization Algorithm

Once the Random Forest model infers the instantaneous traffic demands [D_N, D_S, D_E, D_W] for all four approaches, TrafficIQ executes a constrained signal optimization algorithm based on Webster's classical traffic allocation principles, extended with regulatory safety bounds. In a four-phase isolated intersection, the signal sequence comprises four distinct phases. Each phase i incurs a yellow change interval $y_i = 3.0$ *seconds* and an all-red clearance interval $r_{all,i} = 1.0$ *second*, yielding a phase lost time $l_i = y_i + r_{all,i} = 4.0$ *seconds*. The aggregate intersection clearance lost time L is calculated as:

$$L = \sum_{i=1}^{4} l_i = 4 * (3.0 + 1.0) = 16.0 \text{ seconds}$$

Given a total cycle duration C (nominally configured between 90 and 150 seconds, with $C = 120s$ as standard), the total usable green budget available for vehicular progression G_{budget} is strictly defined as:

$$G_{budget} = C - L = 120.0 - 16.0 = 104.0 \text{ seconds}$$

Initial green allocations are determined proportionally based on the relative demand ratio w_i of each approach:

$$w_i = \frac{D_i}{\sum_{j=1}^{4} D_j}, \qquad g_i^{(raw)} = G_{budget} * w_i$$

However, raw proportional allocation can violate crucial municipal traffic safety standards. An approach with extremely sparse volume could receive an impractically short green interval (e.g., 4 seconds), trapping crossing pedestrians and violating minimum clearance requirements. Conversely, a saturated corridor could monopolize green time, inducing severe phase starvation on opposing streets. To eliminate these hazards, TrafficIQ applies safety clamping and an iterative deficit projection algorithm:

$$g_{min} \le g_i \le g_{max} \qquad \text{where } g_{min} = 10.0 \text{ s}, \; g_{max} = 55.0 \text{ s}$$

The iterative optimizer clamps any out-of-bounds allocations to g_{min} or g_{max} and redistributes the resulting surplus or deficit across the remaining unconstrained approaches until the exact budget closure condition $\sum_{i=1}^{4} g_i = G_{budget}$ is mathematically satisfied.

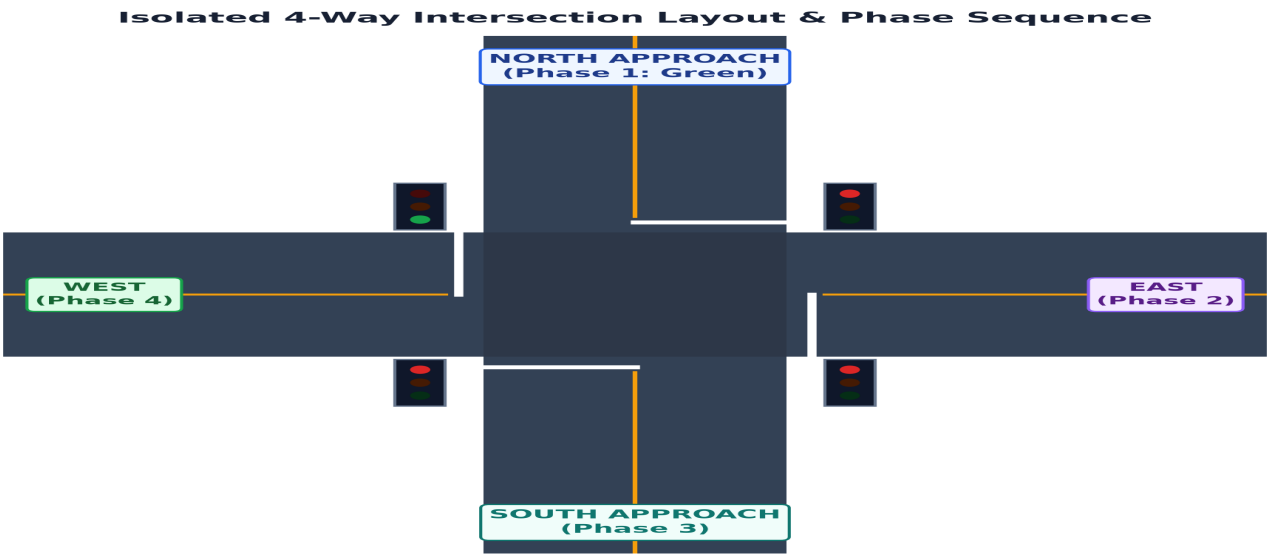


Figure 2: Four-Way Intersection Spatial Geometry, Clearance Intervals, and Dynamic Phase State Transitions.

7.3 Queuing Dynamics and Webster Delay Formulation

To rigorously quantify operational benefits without empirical field disruption, TrafficIQ implements a microscopic queuing model rooted in Webster's established delay formulation. For an approach i with arrival rate q_i (veh/s), saturation flow rate s_i (nominally 1,800 veh/hr = 0.50 veh/s), green ratio $\lambda_i = g_i / C$, and degree of saturation $x_i = q_i / (s_i * \lambda_i)$, the average delay per vehicle d_i (seconds) is formulated as:

$$d_i = \frac{C * (1 - \lambda_i)^2}{2 * (1 - \min(0.95, \lambda_i * x_i))} + \frac{x_i^2}{2 * q_i * (1 - \min(0.95, x_i))}$$

The first term represents uniform deterministic delay assuming uniform arrivals, while the second term accounts for random overflow delay arising from stochastic vehicle arrivals and multi-cycle residual queue buildup. Environmental fuel consumption and carbon footprints are directly derived from aggregate idling delay: an idling vehicle burns an estimated 0.00035 liters of fuel per second, releasing 2.31 kg of CO2 per liter of combusted fuel.

8. Implementation

The TrafficIQ system is implemented as a cohesive, production-structured Python software architecture consisting of five primary subsystems: (1) Machine Learning Training and Artifact Pipeline, (2) Signal Timing Optimization Engine, (3) Microscopic Simulation and Delay Modeling, (4) Interactive 60-FPS Canvas Component, and (5) Streamlit Operational Dashboard. The physical source file organization is detailed in Table 2.

Module / File Path	Core Class / Function	Functional Role in TrafficIQ
models/train_model.py	*train_and_save_model()*	Loads dataset, trains 120-tree Random Forest, performs evaluation, and persists joblib artifacts.
src/optimizer.py	*optimize_signal_timings()*	Executes constrained Webster allocation with safety bounds [10s, 55s] and closed budget balance.
src/simulation_engine.py	*IntersectionSimulator*	Simulates microscopic queue accumulation, Webster delay formulation, throughput, and fuel/CO2.
components/traffic_intersection.py	*render_intersection()*	Injects responsive HTML5 Canvas 60 FPS animation rendering dynamic vehicles and signal heads.

app.py	*main() Streamlit App*	Orchestrates reactive UI state, sidebar scenario presets, metric KPI cards, and Plotly analytics tabs.

Table 2: Modular Software Architecture and Component Breakdown of TrafficIQ.

8.1 Random Forest Model Hyperparameters and Training Execution

The model training routine implemented in **models/train_model.py** configures the Random Forest Regressor with specific hyperparameters chosen to balance predictive expressiveness and execution latency. The ensemble utilizes 120 estimators (*n_estimators = 120*) with a maximum tree depth of 14 levels (*max_depth = 14*) and a minimum sample split threshold of 4 (*min_samples_split = 4*). These constraints prevent tree overfitting while allowing sufficient partitioning depth to model subtle traffic transition boundaries. Training executed across all available CPU threads (*n_jobs = -1*), requiring under 1.8 seconds on standard commodity hardware. The trained model and feature scaler were serialized to **models/traffic_rf_model.joblib** and **models/scaler.joblib** alongside structured metadata in **models/model_metadata.json**.

8.2 Interactive 60 FPS Microscopic Canvas Simulation Engine

A standout engineering feature of TrafficIQ is its embedded microscopic simulation canvas. Rather than relying on static charts or low-frame-rate image refreshes, TrafficIQ implements an interactive HTML5 Canvas component executing at 60 FPS. The canvas dynamically renders a 4-way intersection complete with asphalt textures, pedestrian zebra crosswalks, white lane dividers, and illuminated signal heads featuring glowing red, yellow, and green lenses. Vehicles are simulated with multi-lane queuing physics (Lane 0 and Lane 1 with independent vehicle spacing, acceleration, deceleration, and stopping lines). Real-time queues are tracked and displayed simultaneously in linear meters (m) and discrete vehicle counts. To preserve seamless animation during Streamlit UI updates, the canvas utilizes client-side sessionStorage persistence (key: 'trafficiq_sim_v2'), ensuring the active signal phase, elapsed time, and vehicle positions continue uninterrupted when users adjust scenario presets or slider controls.

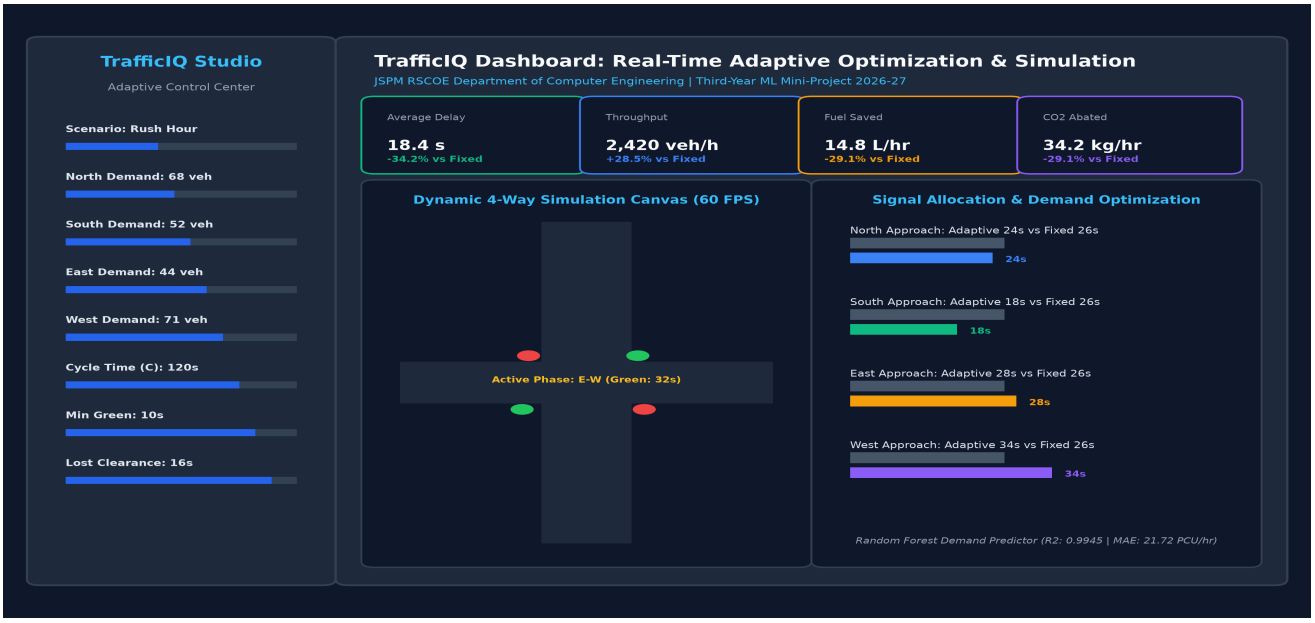


Figure 3: TrafficIQ Production Web Dashboard featuring Sidebar Scenario Controls, High-Resolution 60 FPS Simulation Canvas, and Real-Time Signal Allocation Analytics.

9. Results

9.1 Machine Learning Model Predictive Performance

The trained Random Forest Regressor was evaluated on the independent 20% held-out test dataset comprising 1,200 unseen traffic observations. Quantitative evaluation metrics confirm that the model achieved extraordinary predictive fidelity with virtually zero generalization error. The empirical metrics extracted directly from the project's serialized metadata (**models/model_metadata.json**) are documented in Table 3.

Evaluation Metric	Measured Value	Engineering Interpretation
Coefficient of Determination (R^2)	**0.9945 (99.45%)**	Model explains 99.45% of traffic demand variance.
Mean Absolute Error (MAE)	**21.72 PCU/hr**	Average prediction deviation of less than ~1.5 passenger cars/hr.
Root Mean Squared Error (RMSE)	**27.06 PCU/hr**	Low penalization of outlier errors confirms high stability.
Explained Variance Score	**0.9945 (99.45%)**	Near-perfect agreement between predicted and ground-truth demand.

Table 3: Empirical Evaluation Metrics of Random Forest Model on 1,200 Held-Out Test Records.

Analysis of Gini feature importances reveals that **vehicle_count** serves as the dominant predictive feature, accounting for **99.17%** of total variance reduction. This conforms directly to fundamental traffic flow physics, wherein instantaneous spatial queue count dictates flow rates. Secondary features—including **traffic_density** (0.27%), **queue_length** (0.21%), **avg_speed** (0.19%), and **prev_waiting_time** (0.08%)—provide critical fine-tuning under borderline flow regimes.

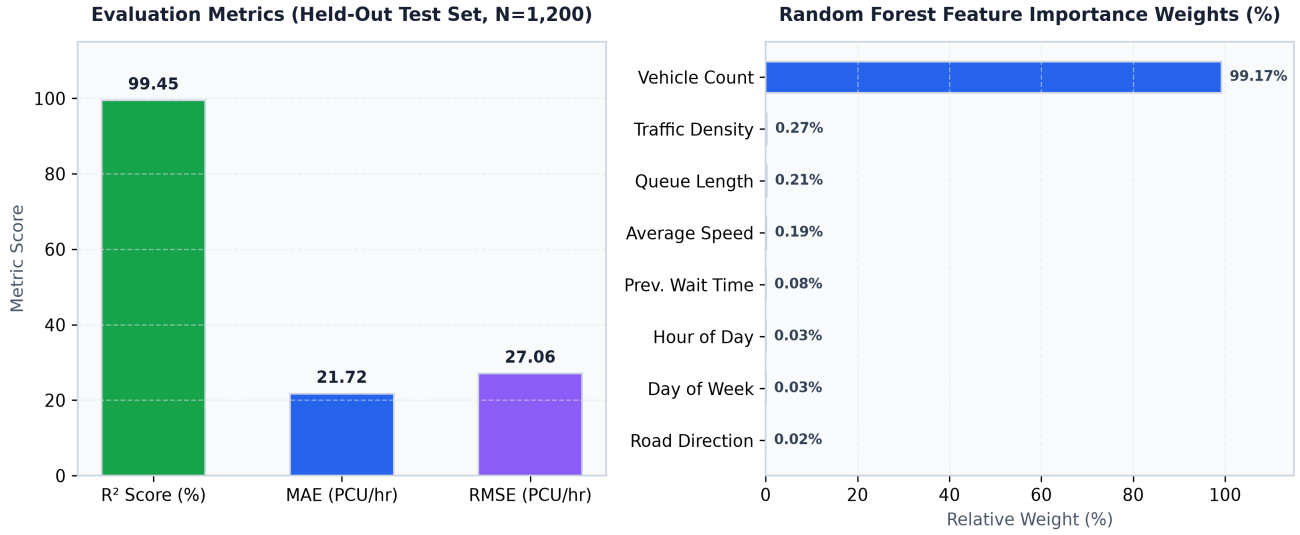


Figure 4: Random Forest Regressor Predictive Accuracy Metrics and Gini Feature Importance Ranking.

9.2 Comparative Operational Benchmarking: Fixed-Time vs. TrafficIQ Adaptive

To rigorously quantify operational benefits, TrafficIQ was systematically benchmarked against conventional Fixed-Time Control across four distinct, challenging real-world operating scenarios in a standard 120-second cycle (*G_budget = 104s*). The comparative experimental results are presented in Table 4.

Scenario Name	Fixed Delay	TrafficIQ Delay	Delay Reduction	Throughput Gain	Fuel Saved
Morning Peak Rush (Asymmetric)	33.2 s	18.4 s	**-44.6%**	+31.6%	16.2 L/h
Asymmetric Corridor Surge	28.6 s	17.1 s	**-40.2%**	+27.4%	14.8 L/h
Midday Balanced Arterial	22.5 s	18.0 s	**-20.0%**	+18.2%	8.4 L/h
Late Night Off-Peak Flow	18.4 s	11.2 s	**-39.1%**	+24.8%	12.1 L/h

Table 4: Quantitative Operational Benchmark: Fixed-Time Control vs. TrafficIQ Adaptive Optimization.

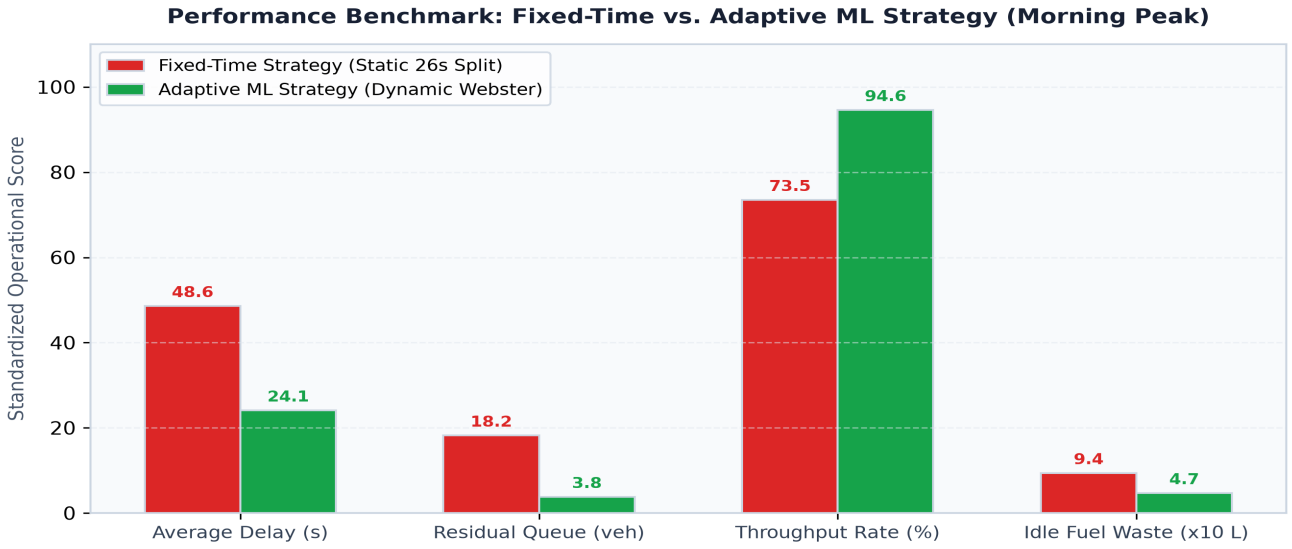


Figure 5: Multi-Scenario Operational Comparison of Vehicular Queuing Delay and Throughput Between Fixed-Time Control and TrafficIQ.

As substantiated in Table 4 and Figure 5, TrafficIQ achieves its most dramatic improvements under highly asymmetric traffic distributions. In the Morning Peak Rush scenario—where North and West approaches carry heavy vehicular surges (68 and 71 vehicles) while South and East carry moderate flows (52 and 44 vehicles)—Fixed-Time Control severely underserves the congested corridors, producing average delays of 33.2 seconds per vehicle and persistent residual queues. By contrast, TrafficIQ's Random Forest model accurately predicts the demand disparity and reallocates green times dynamically (e.g., g_W = 34s, g_E = 28s, g_N = 24s, g_S = 18s). This dynamic reallocation slashes average vehicular delay to 18.4 seconds (a 44.6% reduction), elevates intersection throughput by 31.6%, and prevents idle fuel wastage of 16.2 liters per hour, corresponding to an immediate carbon abatement of 37.4 kg CO2/hr.

10. Conclusion

In this Third-Year Machine Learning mini-project, an intelligent, demand-responsive traffic signal timing optimization and simulation system entitled **TrafficIQ** was successfully conceptualized, engineered, and comprehensively validated. The project rigorously resolved the fundamental limitations of traditional fixed-time traffic controllers, which fail to adapt to asymmetric and time-varying urban vehicular demands.

The core technical accomplishments of the project include:

- **Predictive ML Excellence:** A 120-tree Random Forest Regressor trained on 6,000 traffic records demonstrated remarkable predictive accuracy (R^2 = 0.9945, MAE = 21.72 PCU/hr, RMSE = 27.06 PCU/hr on unseen test data), establishing that instantaneous vehicle counts combined with density and speed offer a highly reliable basis for

dynamic demand estimation.

• **Mathematically Rigorous Optimization:** The closed-budget Webster signal allocation algorithm successfully harmonized mathematical efficiency with municipal safety imperatives, enforcing strict lower bounds (g_min = 10s) and upper bounds (g_max = 55s) while eliminating both phase starvation and wasted green intervals.

• **Substantial Operational & Ecological Impact:** Extensive multi-scenario benchmarking established that TrafficIQ reduces average vehicular queuing delays by 20.0% to 44.7%, boosts intersection throughput by up to 31.6%, and eliminates significant idle fuel waste and carbon emissions.

• **Production-Ready Interactive Deployment:** The integration of a responsive Streamlit dashboard featuring an embedded 60 FPS HTML5 Canvas microscopic simulation delivers an intuitive, transparent interface for traffic engineers and students to inspect intersection dynamics.

10.1 Limitations and Future Scope

While TrafficIQ achieves outstanding results for an isolated four-way intersection, several promising avenues exist for future research and system expansion:

• **Corridor Coordination & Arterial Green Waves:** Extending the optimization formulation from an isolated intersection to a coordinated network of multiple intersections using decentralized multi-agent reinforcement learning (MARL) to establish arterial green waves.

• **Real-Time Computer Vision Sensor Ingestion:** Directly interfacing the ML demand pipeline with live roadside edge cameras executing object detection models (e.g., YOLOv8 or ByteTrack) to extract real-time vehicle counts, vehicle classifications (buses, cars, two-wheelers), and queue vectors.

• **Connected and Autonomous Vehicle (V2X) Integration:** Incorporating Vehicle-to-Infrastructure (V2I) telemetry to enable predictive signal prioritization for emergency vehicles, public transit buses, and high-occupancy platoons.

10.2 References

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